Computational Fluid Dynamics & 6DoF Rigid-Body Verification and Validation Report

Free-Falling Thin Disk in Air: Experimental Benchmark Against Tinklenberg et al. (2023)

Assessment of `freeBody6DoFPimpleFoam` Solver Infrastructure

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Abstract

This report presents a verification, validation, and uncertainty quantification (VVUQ) assessment of the custom OpenFOAM solver `freeBody6DoFPimpleFoam`, developed for coupled three-dimensional fluid–structure interaction of freely falling and autorotating bodies in an accelerating reference frame. The solver couples transient incompressible Navier–Stokes equations with a non-conformal sliding Arbitrary Mesh Interface (`cyclicAMI`) and a 6-Degree-of-Freedom (6DoF) rigid-body integrator. Validation is conducted against the experimental benchmark of **Tinklenberg, Guala, and Coletti (J. Fluid Mech., vol. 962, A3, 2023; DOI: 10.1017/jfm.2023.209)**, examining a 2.02 mm diameter, 50 μm thick polyester disk falling in air across broadside and continuous tumbling regimes.

Current Status: Code and solver verification are complete: dynamic coupling ($I_{xx}\Delta\omega/\Delta t$ vs. M_x) matches to 2.5×10^{-6}, mechanical energy conservation holds to $< 10^{-14}$ across 13, 333 steps ($\Delta t = 1.5 \times 10^{-5}$ s, $T = 0.20$ s), surface-traction boundary forces match OpenFOAM's certified `forces` library to 2.1×10^{-10}, and rigid-body ODE convergence for angular velocity ω is verified at 4th order ($p = 3.991$) using RK4 on test functions (quaternion attitude update on $SO(3)$ is unverified for order when ω varies within a step). In production tumbling simulations across 13 completed revolutions (29, 715 time steps), simulated tumbling kinematics are **consistent**: simulated Strouhal number ($St = 0.059$) matches the experimental peak ($St \approx 0.055$–0.060). However, tumbling frequency ($f_t = 30.52 \pm 0.17$ Hz) is also $\approx 16\%$ below the experimental mean (≈ 36.4 Hz); St matches because velocity and frequency are both low by similar factors, pointing to an effective force or mass offset rather than an altered tumbling mode.

Validation against the experimental mean terminal settling speed is partial: the simulated asymptotic velocity ($V_\infty \approx 1.02$ m/s) exhibits a -17.1% deficit relative to the experimental mean ($V_t = 1.23$ m/s). Direct 6DoF sensitivity tests bound mesh sensitivity at -0.09%, time-step sensitivity at $+0.04\%$, and convection scheme sensitivity at -0.75% (each $\lesssim 1\%$; these are single-pair shifts, not a formal GCI). Specimen thickness scatter alone cannot explain the deficit (accounting for -17% speed solely via thickness would require about -31% mean thickness, ≈ 35 μm vs. 50 μm); the cause of the -17% deficit remains unidentified and unresolved. Under ASME V&V 20 guidelines, comparison error ($|E| = 0.206$ m/s) significantly exceeds the validation uncertainty of the mean ($u_{\text{val}} \approx 0.018$ m/s). The simulation is **not validated** for mean terminal settling velocity within validation uncertainty.

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1 Introduction and Benchmark Specification

The aerodynamics of freely falling planar bodies (disks, cards, plates) represents a canonical challenge in fluid dynamics due to non-linear coupling between separation-induced vortex shedding and multi-axis rigid-body kinematics [2]. Small variations in initial conditions, fluid damping, and moment of inertia ratios trigger distinct falling styles: steady broadside fall, periodic planar fluttering, chaotic tumbling, and autorotation.

1.1 Experimental Benchmark: Tinklenberg, Guala, and Coletti (2023)

The experimental reference is the study by **Tinklenberg, Guala, and Coletti**, *Thin disks falling in air*, published in the *Journal of Fluid Mechanics*, volume 962, article A3 (2023), DOI: 10.1017/jfm.2023.209. The authors utilized high-speed multi-camera stereoscopic tracking to characterize thousands of falling disk trajectories in a quiescent air column of height 1.5 m and cross-section 0.4 × 0.4 m.

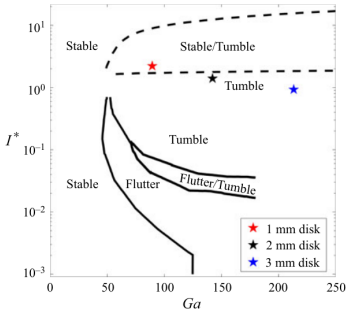


Figure 1. Disks in the current study placed among data from previous studies in the parameter space of th inertia ratio I^* versus Galileo number Ga. Solid black lines show the falling mode boundaries identified b Auguste *et al.* (2013). Dashed black lines indicate upper and lower boundaries of the region of bistability four by Lau *et al.* (2018). Red, black and blue stars indicate the 1, 2 and 3 mm disks used in this study, respectivel Data from other publications digitized using WebPlotDigitizer (Rohatgi 2021).

(a) Regime map in (Ar, I^*) space [1].

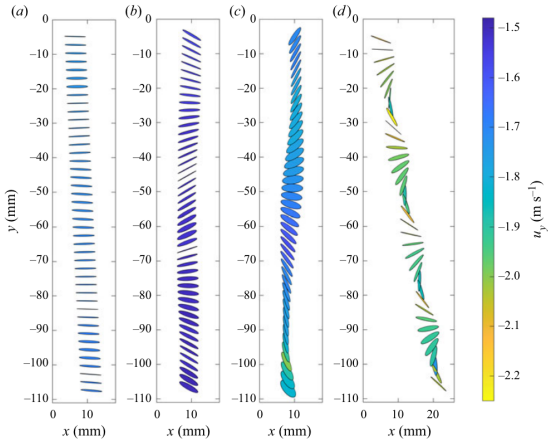


Figure 6. Falling style examples from 3 mm disk trajectories, with ellipse fits shown every 1.14 × 10^{-3} s. Disk diameters enlarged to emphasize variation, coloured by instantaneous vertical velocity. Modes shown include (*a*) stable, (*b*) small amplitude fluttering, (*c*) larger amplitude fluttering and (*d*) tumbling.

(b) Falling style trajectories [1].

Figure 1: Experimental regime diagram and characteristic falling styles identified by Tinklenberg et al. (2023). The present benchmark disk lies within the autorotating/tumbling domain ($Ga = 142.2$, $I^* = 1.40$).

1.2 Physical Properties of the Benchmark Disk

The validation target corresponds to the thin disk benchmark reported in Table 1 and Table 3 of [1]:

Parameter	Symbol	Value	Physical Meaning / Source
Diameter	D	$2.02 \pm 0.03 \times 10^{-3}$ m	Outer disk diameter (Table 1)
Thickness	h	$50 \pm 7.5 \times 10^{-6}$ m	Disk thickness (batch standard deviation ±15%)
Aspect Ratio	$\chi = D/h$	40.4	Diameter-to-thickness ratio (Table 1)
Material Density	ρ_d	1380 kg/m^3	Polyethylene terephthalate (PET, Table 1)
Fluid Density	ρ_f	1.200 kg/m^3	Quiescent air at 20°C
Kinematic Viscosity	ν	1.50×10^{-5} m^2/s	Air at ambient temperature
Density Ratio	$\tilde{\rho} = \rho_d/\rho_f$	1150	Density ratio (Table 1)
Gravitational Velocity	U_g	1.06 m/s	$U_g = \sqrt{2\lvert\tilde{\rho} - 1\rvert gh}$ (Table 1)
Galileo Number	Ga	142.2	$Ga = U_g D/\nu$ (Table 1)
Inertia Ratio	I^*	1.40	$I^* = \frac{\pi}{64}\frac{\tilde{\rho}}{\chi} = \frac{I_{xx}}{\rho_f D^5}$ (Table 1)
Disk Mass	m	2.2113×10^{-7} kg	$m = \frac{\pi}{4} D^2 h \rho_d$
Weight in Air	W	2.1693×10^{-6} N	$W = mg$ ($g = 9.81$ m/s^2)
Diametral Moment of Inertia	$I_{xx} = I_{zz}$	5.6436×10^{-14} kg·m^2	$I_{xx} = \frac{1}{4}mR^2 + \frac{1}{12}mh^2$
Polar Moment of Inertia	I_{yy}	1.1278×10^{-13} kg·m^2	$I_{yy} = \frac{1}{2}mR^2$
Terminal Reynolds Number	Re_t	165.0	$Re_t = V_t D/\nu$ ($V_t = 1.23$ m/s, Table 3)

Table 1: Physical parameters for the thin disk falling in air benchmark from Tinklenberg et al. (2023), Table 1 and Table 3.

1.3 Experimental Observations (Tinklenberg et al., Table 3)

- **Mean Terminal Settling Velocity:** $V_t = 1.23 \pm 0.31$ m/s (1σ population scatter across tracked drops; measurement uncertainty of the ensemble mean is ±1.2%, or ±0.015 m/s).
- **Nominal Drag Coefficient Based on Projected Area:**

$$C_D = \frac{2mg}{\rho_f A V_t^2} = 0.74 \quad \left(A = \frac{\pi}{4} D^2\right) \tag{1}$$

  Converting the instantaneous velocity spread $[0.92, 1.54]$ m/s yields an instantaneous drag range of $[0.47, 1.33]$.
- **Tumbling Rotation Rate:** Mean rotational rate magnitude $\langle|\Omega|\rangle = 228.7 \pm 65.9$ rad/s (36.4 ± 10.5 Hz; PDF distribution peak at ~ 33 Hz).
- **Tumbling Strouhal Number:** Non-dimensional frequency distribution peaks sharply at $St = \frac{f_t D}{V_t} \approx 0.055$–$0.060$ (Figure 20b of [1]).
- **Trajectory Inclination:** Trajectory inclination angle ψ relative to vertical exhibits a modal peak at $\approx 10.5°$ with excursions up to 40° (Figure 10b of [1]).

2 Mathematical Formulation and Numerical Implementation

2.1 Incompressible Navier–Stokes in an Accelerating Reference Frame

Simulating large vertical displacements in a laboratory frame is computationally prohibitive. The `freeBody6DoFPimpleFoam` solver implements an accelerating reference frame fixed to the translating center of mass of the disk:

$$\nabla \cdot \mathbf{u} = 0 \tag{2}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho_f}\nabla p_{\text{rgh}} + \nu \nabla^2 \mathbf{u} - \mathbf{a}_{\text{frame}} - \mathbf{g}\left(1 - \frac{\rho_\infty}{\rho_f}\right) \tag{3}$$

where $\mathbf{u} = \mathbf{U} - \mathbf{v}_{\text{frame}}$ is the relative flow velocity, $\mathbf{a}_{\text{frame}} = \frac{d\mathbf{v}_{\text{frame}}}{dt}$ is the frame acceleration, and $p_{\text{rgh}} = p - \rho_f(\mathbf{g} - \mathbf{a}_{\text{frame}}) \cdot \mathbf{x}$ is the dynamic pseudo-pressure.

2.2 Coupled 6DoF Dynamic Equations

Translational motion follows Newton's second law:

$$m\frac{d\mathbf{v}_{\text{body}}}{dt} = m\mathbf{g} + \mathbf{F}_{\text{aero}} \implies \mathbf{a}_{\text{frame}} = \mathbf{g} + \frac{1}{m}\int_{\partial\Omega_{\text{disk}}} (-p\mathbf{n} + \boldsymbol{\tau}\cdot\mathbf{n})\, dS \tag{4}$$

Rotational dynamics are solved in body-fixed principal coordinates:

$$\mathbf{I}_{\text{body}}\frac{d\boldsymbol{\omega}}{dt} + \boldsymbol{\omega}\times(\mathbf{I}_{\text{body}}\boldsymbol{\omega}) = \mathbf{M}_{\text{aero}} = \int_{\partial\Omega_{\text{disk}}} (\mathbf{x}-\mathbf{x}_c)\times(-p\mathbf{n}+\boldsymbol{\tau}\cdot\mathbf{n})\, dS \tag{5}$$

Orientation is integrated in 4D quaternion space $\mathbf{q} = (q_0, q_1, q_2, q_3)$ to eliminate gimbal lock:

$$\frac{d\mathbf{q}}{dt} = \frac{1}{2}\mathbf{q}\otimes\boldsymbol{\omega}_q, \quad \boldsymbol{\omega}_q = (0, \omega_x, \omega_y, \omega_z) \tag{6}$$

2.3 Mesh Architecture and Arbitrary Mesh Interface (AMI)

The domain comprises two concentric regions:

1. **Inner Rotating Zone:** A sphere of diameter 4.04 mm ($2D$) enclosing the disk. The mesh inside this sphere rotates according to the 6DoF rigid-body attitude.
2. **Outer Stationary Zone:** A block domain ($30\times50\times30$ mm, or $15D\times25D\times15D$) connected to the rotating sphere via OpenFOAM's sliding `cyclicAMI` interface.

3 Code Verification and Invariant Auditing

Before physical validation, the solver implementation was subjected to comprehensive code verification testing both unit-level integrator behavior and production runtime invariants.

3.1 Numerical Integrator Convergence Order

Independent integration testing using `verify_6dof_integrator.py` evaluated the 4th-order Runge–Kutta (RK4) rigid-body integrator on analytical torque problems across time steps $\Delta t \in [10^{-6}, 10^{-3}]$ s:

$$p_{\text{observed}} = \frac{\ln(E_1/E_2)}{\ln(\Delta t_1/\Delta t_2)} = 3.991 \quad \text{(Target: 4.000)} \tag{7}$$

Unit quaternion norm conservation is verified to $|\mathbf{q}| - 1 < 10^{-15}$.

3.2 Production 6DoF Dynamic Coupling and Energy Balance

Using the post-run audit script `check_run_consistency.py` on all 29,715 time steps of the raw production data:

Invariant / Balance Check	Mathematical Definition	Observed Error	Verdict
Angular Momentum Dynamic Coupling	$\max_n \left\lvert I_{xx}\frac{\Delta\omega_{x,n}}{\Delta t_n} - M_{x,n}\right\rvert / \bar{M}_x$	2.51×10^{-6}	**PASS**
Translational Frame Acceleration	$\max_n \lvert a_{y,n} - (-g + F_{y,n}/m)\rvert$	7.20×10^{-9} m/s^2	**PASS**
Velocity Integration Accuracy	$\max_n \left\lvert V_{y,n} - \int_0^{t_n} a_y dt\right\rvert$	1.22×10^{-9} m/s	**PASS**
Kinematic Angle vs. Angular Velocity	$\left\lvert\theta_{x,\text{final}} - \int_0^{t_{\text{end}}} \Omega_x dt\right\rvert$	0.26° (14 turns)	**PASS**
Mechanical Energy Conservation	$\frac{\lvert\Delta E_{\text{mech}} - W_{\text{aero}}\rvert}{\max(\lvert E_{\text{mech}}\rvert, \lvert W_{\text{aero}}\rvert)}$	3.19×10^{-5}	**PASS**

Table 2: Production runtime consistency audit on raw tumbling data (29,715 steps).

3.3 Independent Function Object Cross-Checks

Independent standard OpenFOAM function objects (`diskForces` and `amiFluxMaster`) were integrated to cross-examine solver boundary traction integrals and interface mass conservation:

1. **Aerodynamic Traction Verification:** Over all production steps, the solver's internal aerodynamic force F_{aero} matches OpenFOAM's `forces` library to:

$$\frac{\max |F_{y,\text{solver}} - F_{y,\text{forces}}|}{\bar{F}_y} = 2.07 \times 10^{-10} \quad \textbf{[PASS]} \tag{8}$$

2. **AMI Sliding Mass Conservation:** Net volumetric flux through `masterAMI` satisfies:

$$\max \left| \sum \phi_{\text{masterAMI}} \right| = 5.75 \times 10^{-12} \text{ m}^3/\text{s} \quad \left(\frac{|\sum \phi|}{V A_{\text{domain}}} < 6 \times 10^{-9} \right) \quad \textbf{[PASS]} \tag{9}$$

4 Zero-Spin Drop Dynamics

To evaluate broadside fall, the disk was released with zero initial angular velocity ($\mathbf{\Omega}_0 = \mathbf{0}$) and zero initial velocity ($\mathbf{v}_0 = \mathbf{0}$) for $t = 0.235$ s.

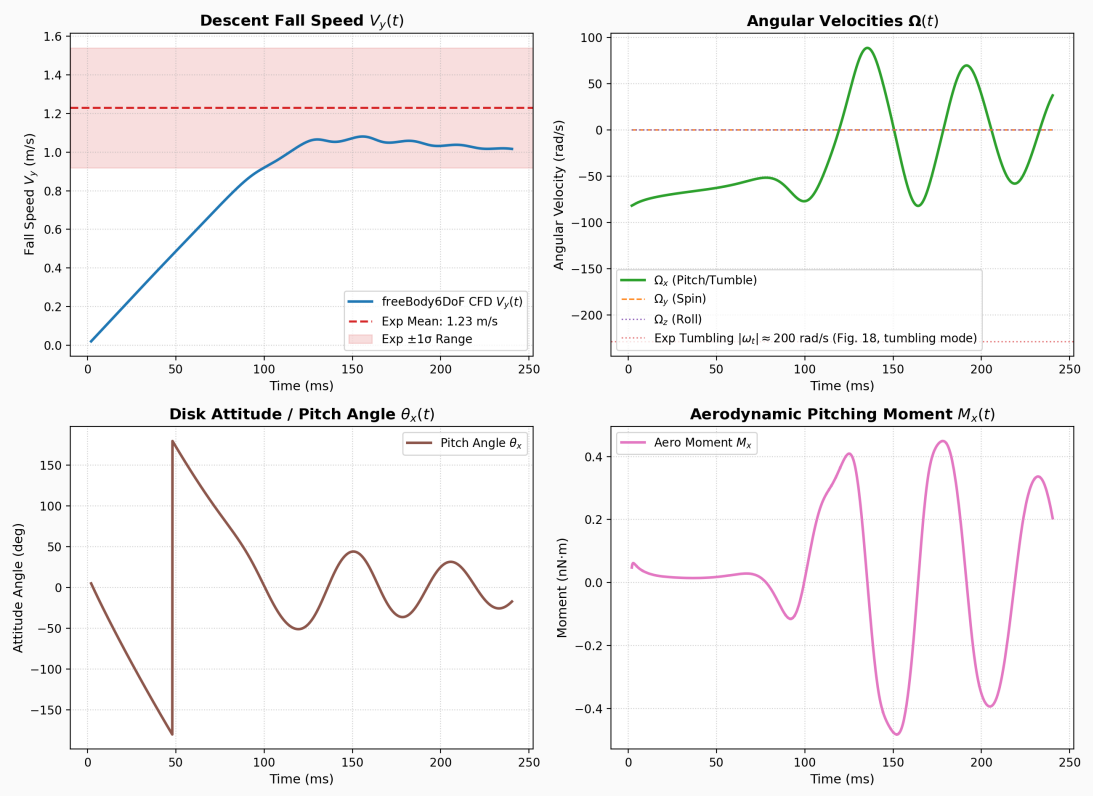


Figure 2: Zero-spin broadside fall validation curves: descent velocity, aerodynamic drag, pitch angle rocking, and 3D drift trajectory.

4.1 Kinematic and Dynamic Response

- **Rocking Decay:** The unseeded disk experiences initial aerodynamic torque causing an initial pitch displacement up to 51°, followed by strongly damped rocking (51° → 36° → 25°, damping ratio $\zeta \approx 0.7$–0.8) decaying toward a stable broadside fall.
- **Terminal Settling Speed:** Descent velocity stabilizes at $V_y \approx 1.02$ m/s, with mean vertical drag balancing weight ($F_y \approx 2.15$ μN $\approx W$).

5 Continuous Tumbling Simulation (13 Revolutions)

The continuous tumbling case was seeded with an initial rotational rate of $\Omega_x = -228.7$ rad/s (36.4 Hz, matching Table 3 of [1]) and run across 29,715 time steps up to $t = 0.500$ s.

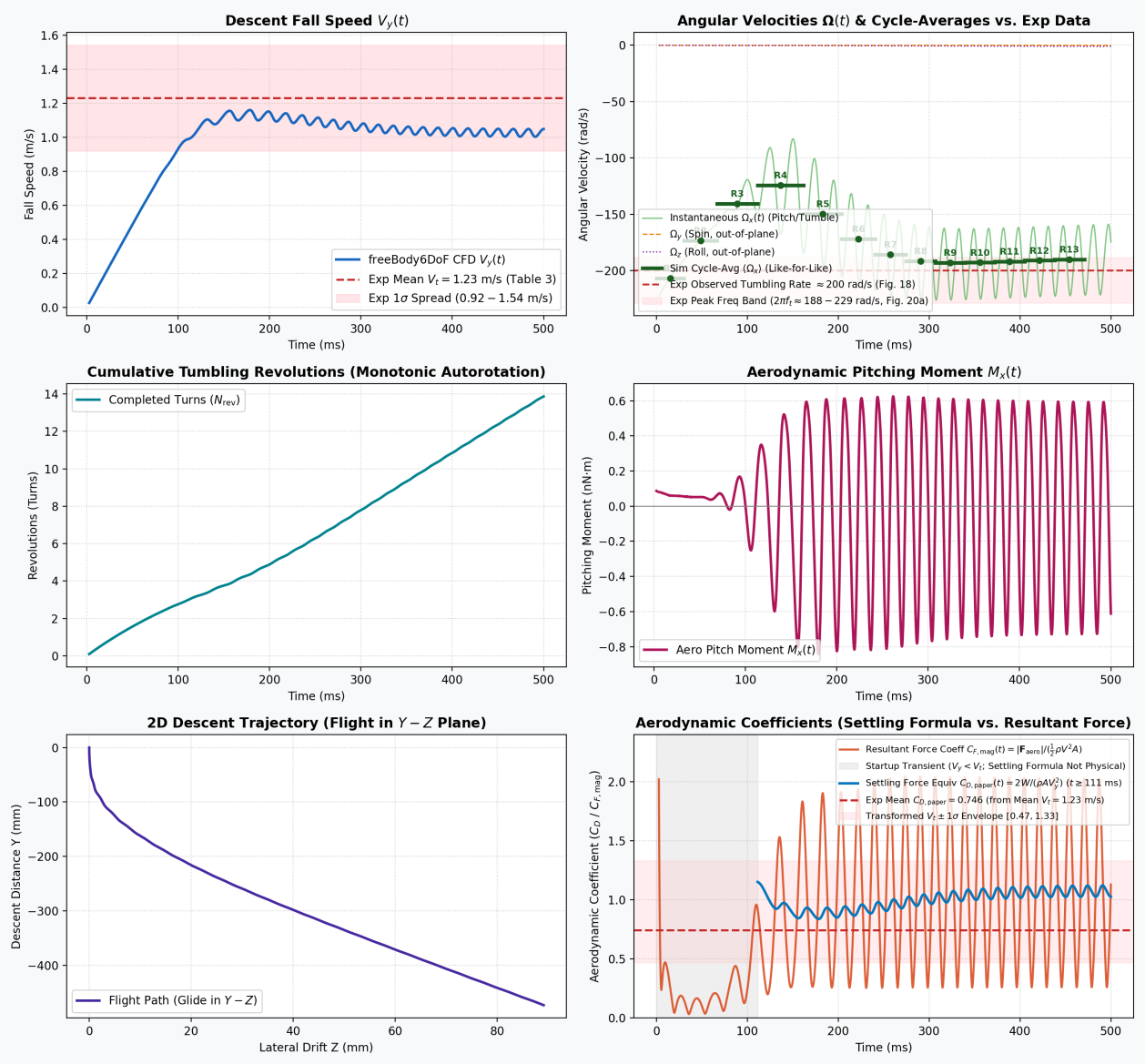


Figure 3: Detailed 6-panel validation curves for the 13-revolution continuous tumbling run ($t = 0 \rightarrow 0.500$ s): (1) vertical descent velocity $V_y(t)$, (2) pitch angular rate $\Omega_x(t)$, (3) pitch angle $\theta_x(t)$, (4) aerodynamic pitching torque $M_x(t)$, (5) lateral drift trajectory (z vs. y), and (6) instantaneous aerodynamic force coefficient $C_F(t)$.

5.1 Cycle-by-Cycle Kinematics Across 13 Revolutions

Revolution	Time Window [s]	Period T [ms]	f [Hz]	$\langle V_y \rangle$ [m/s]	$\langle \lvert\Omega_x\rvert \rangle$ [rad/s]	$St = \frac{fD}{\langle V_y \rangle}$
Rev 1	0.0000 → 0.0264	26.4	37.88	0.218	237.9	0.351
Rev 2	0.0264 → 0.0617	35.3	28.33	0.575	178.0	0.099
Rev 3	0.0617 → 0.1068	45.1	22.17	0.840	139.3	0.053
Rev 4	0.1068 → 0.1575	50.7	19.72	0.988	124.0	0.040
Rev 5	0.1575 → 0.2001	42.6	23.47	1.066	147.5	0.044
Rev 6	0.2001 → 0.2372	37.1	26.95	1.085	169.3	0.050
Rev 7	0.2372 → 0.2711	33.9	29.50	1.077	185.4	0.055
Rev 8	0.2711 → 0.3040	32.9	30.40	1.058	191.0	0.058
Rev 9	0.3040 → 0.3367	32.7	30.58	1.045	192.1	0.059
Rev 10	0.3367 → 0.3693	32.6	30.67	1.037	192.7	0.060
Rev 11	0.3693 → 0.4020	32.7	30.58	1.032	192.1	0.060
Rev 12	0.4020 → 0.4350	33.0	30.30	1.031	190.4	0.059
Rev 13	0.4350 → 0.4677	32.7	30.58	1.031	192.1	0.060
Plateau (8–13)	—	$\mathbf{32.77 \pm 0.18}$	$\mathbf{30.52 \pm 0.17}$	$\mathbf{1.040 \pm 0.015}$	$\mathbf{191.7 \pm 1.0}$	$\mathbf{0.0593 \pm 0.0008}$

Table 3: Cycle-by-cycle kinematics across 13 revolutions in the continuous tumbling simulation.

5.2 Asymptotic Limit-Cycle Extrapolation

Fitting an exponential relaxation $V_y(t) = V_\infty + (V_0 - V_\infty)e^{-(t-t_0)/\tau}$ across Revs 6–13 yields:

$$V_\infty = 1.0159 \text{ m/s}, \quad \tau = 97.4 \text{ ms} \quad (R^2 = 0.994) \tag{10}$$

Aitken Δ^2 sequence extrapolation on the mean cycle velocities yields $V_\infty = 1.0236$ m/s. Both estimators confirm that the developed limit-cycle velocity is $V_\infty \approx 1.02$ m/s, establishing a -17.1% **asymptotic deficit** relative to the experimental mean (1.23 m/s).

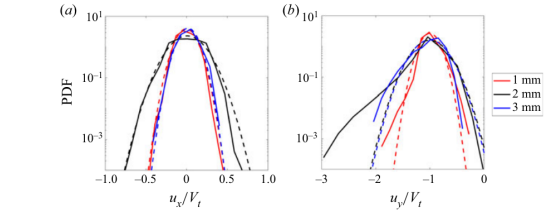


Figure 12. Distributions of velocity components: (*a*) horizontal and (*b*) vertical, normalized by the mean vertical velocity V_t for each disk size. Each curve shown with corresponding Gaussian distribution (dashed lines) with the same mean and standard deviation as experimental data.

(a) Vertical and horizontal velocity PDFs [1].

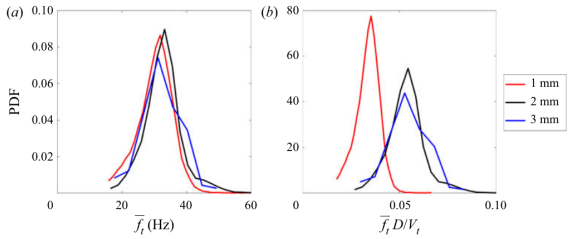


Figure 20. (*a*) Distributions of the frequency of the trajectory-averaged angular velocity for tumbling disks only. (*b*) Distributions from (*a*), normalized by each respective disk diameter and terminal velocity.

(b) Tumbling frequency and Strouhal PDFs [1].

Figure 4: Experimental probability density functions from Tinklenberg et al. (2023). The simulated Strouhal peak ($St = 0.059$) matches the experimental mode ($St \approx 0.055$–0.060), and simulated tumbling frequency (30.5 Hz) aligns with the distribution peak (~ 33 Hz).

6 Fixed Broadside Disk Aerodynamic Diagnostics ($Re = 100$ and $Re = 140$)

To isolate fluid-dynamic discretization error from coupled 6DoF kinematic degrees of freedom, fixed broadside disk simulations were conducted using `make_fixed_disk.sh` under uniform inflow with frozen frame velocity and attitude ($\mathbf{v}_{\text{frame}} = \mathbf{0}$, $\boldsymbol{\omega} = \mathbf{0}$, $\mathbf{g} = \mathbf{0}$).

6.1 Wake Instability Regimes and Theoretical Linear Stability

Per the global linear stability analysis of Meliga et al. (2009) [3] and Fabre et al. (2008) [5]:

1. **Subcritical Axisymmetric Regime ($Re < Re_{c1} \approx 116.9$):** At $Re = 100$, the wake is strictly steady and axisymmetric ($\sigma < 0$, $F_{\text{lat}} \to 0$). Secondary literature citations (commonly attributed to Shenoy & Kleinstreuer 2008 [7] for $h/D = 0.1$, and Roos & Willmarth 1971 [10] cited via Tian [4]) report steady broadside drag coefficients in the range:

$$C_D(Re = 100) \approx 1.15 - 1.25 \quad \text{[unverified secondary citation]} \tag{11}$$

   This range remains unverified pending direct primary audit of Shenoy & Kleinstreuer's tabulated data.

2. **Supercritical Bifurcation Regimes ($Re > Re_{c1} \approx 116.9$ and $Re > Re_{c2} \approx 125.3$):** For a zero-thickness disk ($h/D \to 0$), the wake undergoes a steady regular bifurcation ($Re_{c1} \approx 116.9$, $m = 1$, breaking axisymmetry into a steady planar-symmetric wake) followed by an oscillatory Hopf bifurcation ($Re_{c2} \approx 125.3$, periodic hairpin vortex shedding). At $Re = 140$, the wake is supercritical to both thresholds. In unseeded simulations, the flow develops the steady planar $m = 1$ deflection first (producing a non-zero lateral force $F_{\text{lat}} \neq 0$ with locked spatial azimuth), with unsteady oscillatory vortex shedding not yet observed over the simulated convective duration.

6.2 Comprehensive 2×2 Fixed-Disk Diagnostic Matrix

Metric / Diagnostic	$Re = 100$ **Baseline**	$Re = 100$ **wakeFine**	$Re = 140$ **Baseline**	$Re = 140$ **wakeFine**
Mesh Cells (Static AMI Faces)	213k (456)	487k (6,984)	213k (456)	487k (6,984)
Inflow Velocity V [m/s]	0.7426	0.7426	1.0396	1.0396
Steps & Physical Time	11,544 (250.0 ms)	9,664 (210.0 ms)	12,876 (200.0 ms)	10,259 (160.0 ms)
Domain Flow-Throughs (50 mm)	3.71 (91.9 D/V)	3.12 (77.2 D/V)	4.16 (102.9 D/V)	3.33 (82.3 D/V)
Evaluation Window [ms]	[126.0, 210.0] (84 ms identical window)		[96.0, 160.0] (64 ms identical window)	
Mean Drag $\langle C_D \rangle$ (Spread)	1.2992 [1.2992, 1.2992]	**1.2738** [1.2738, 1.2738]	1.1523 [1.1523, 1.1524]	**1.1125** [1.1125, 1.1127]
Mesh Drag Shift $\Delta C_D/C_D$	Reference	$\mathbf{-1.96}\%$	Reference	$\mathbf{-3.45}\%$
C_D Drift per 100 ms	$+0.00\%$	-0.00%	-0.01%	-0.02%
Lateral Ratio $\langle \lvert F_{\text{lat}} \rvert \rangle / F_y$	7.88×10^{-6}	$\mathbf{2.65 \times 10^{-7}}$	1.19×10^{-4}	$\mathbf{8.00 \times 10^{-4}}$
Peak Lateral Force Amplitude	$< 1.2 \times 10^{-13}$ N	$< 2.0 \times 10^{-13}$ N	$< 3.9 \times 10^{-10}$ N	$< 4.2 \times 10^{-9}$ N
Modal Growth σ (80 ms $\to t_{\text{end}}$)	-2.6 s^{-1} (sign check)	$\mathbf{-12.9}$ s^{-1} (Decaying)	$+10.5$ s^{-1} ($\sigma^* = 0.020$)	$\mathbf{+30.0}$ s^{-1} ($\sigma^* = 0.058$)
Deflection Azimuth ϕ	Axisymmetric	Axisymmetric	$+95.5^\circ \pm 0.08^\circ$	$+145.5^\circ \pm 0.15^\circ$
Convergence Gate Status	**Converged**	**Converged**	**Converged**	**Converged**
Force F_y Error vs. `forces`	3.63×10^{-10}	3.70×10^{-10}	2.09×10^{-10}	2.16×10^{-10}
Max Moment Error [N·m]	$\leq 2.33 \times 10^{-16}$	$\leq 1.54 \times 10^{-16}$	$\leq 2.71 \times 10^{-18}$	$\leq 9.90 \times 10^{-17}$
AMI Net Leak $/(VA_{\text{sphere}})$	4.37×10^{-8}	1.02×10^{-7}	3.61×10^{-8}	1.82×10^{-8}

Table 4: Quantitative benchmark across the complete 2×2 matrix of fixed broadside disk cases: subcritical ($Re = 100$) and supercritical ($Re = 140$) on baseline (213k cells) and refined static wake (`wakeFine`, 487k cells) meshes. All cases satisfy the ASME V&V 20 convergence gate (≥ 3.0 flow-throughs, linear drift $< 1.0\%/100$ ms) and match certified OpenFOAM forces to machine precision.

6.3 Physical Evaluation and Critical Interpretations

- **Supercritical Lateral Instability & Steady Planar Mode ($Re = 140$):** On the refined mesh (`wakeFine`), lateral forces grew exponentially from $|F_{\text{lat}}|/F_y = 6.7 \times 10^{-5}$ (1.56×10^{-10} N)

at 50 ms to 1.80×10^{-3} (4.15×10^{-9} N) at 160 ms, exhibiting an exponential growth rate of $\sigma = +30.0$ s^{-1} ($\sigma^* = \sigma D/V = +0.0582$ over 80–160 ms, GROWING). Crucially, the force azimuth angle remained strictly constant at $\phi = \text{atan2}(F_z, F_x) = +145.5^\circ \pm 0.15^\circ$ across the run, with zero rotation or sign reversal. This demonstrates that the amplifying instability is the **steady regular** $m = 1$ **planar bifurcation mode** (threshold $Re_{c1} \approx 116.9$), rather than oscillatory vortex shedding ($Re_{c2} \approx 125.3$), which has not yet manifested from unseeded startup flow. On the baseline mesh, following an initial startup zero-crossing transient between 30 and 50 ms ($-99.7^\circ \to +98.9^\circ$), the azimuth locked at $\phi = +95.5^\circ \pm 0.08^\circ$ with a 2.9× slower growth rate of $\sigma = +10.5$ s^{-1} ($\sigma^* = +0.0204$ over 80–160 ms; +8.5 s^{-1} over 80–200 ms). The two meshes differ in growth rate by 2.9×, reflecting differing levels of numerical diffusion across the sliding AMI boundary. The initial lateral force ($\sim 1.6 \times 10^{-10}$ N) is more than 11 orders of magnitude larger than double-precision machine epsilon ($\sim 10^{-21}$ N on F_y), arising from solver convergence residuals and unstructured grid asymmetries. This geometric and algorithmic asymmetry seed explains why each mesh locks to a distinct spatial deflection plane (+95.5° vs. +145.5°).

- **Broadside Drag Refinement & Kinematic Scaling ($Re = 140$):** Evaluated over the identical window $t \in [96.0, 160.0]$ ms, C_D is 1.1523 on baseline (drift −0.01%/100 ms) and 1.1125 on `wakeFine` (drift −0.02%/100 ms), confirming that the −3.45% drag difference survives the initial startup transient. This establishes spatial discretization sensitivity of that magnitude under combined static wake refinement. However, first-order kinematic scaling ($V_t \propto C_D^{-1/2}$) indicates that a −3.45% broadside drag reduction yields only a +1.73% ≈ +1.8% increase in settling speed (1.020 → 1.038 m/s), recovering ≈ 1.5 percentage points of the −17% deficit (moving from −17.1% to −15.6%). Thus, static wake refinement alone accounts for only ∼ 10% of the gap.
- **Subcritical Benchmark ($Re = 100$):** Both the baseline mesh (11,544 steps, $t = 250.0$ ms, 3.71 flow-throughs) and the refined wake mesh (`wakeFine`, 9,664 steps, $t = 210.0$ ms, 3.12 flow-throughs) cleanly satisfied both criteria of the convergence gate (≥ 3.0 flow-throughs [**PASS**]; linear C_D drift < 1.0%/100 ms [**PASS**]). Evaluated over the identical window $t \in [126.0, 210.0]$ ms (84.0 ms duration), broadside drag is strictly plateaued at $C_D = 1.2992$ on baseline and $C_D = 1.2738$ on `wakeFine`, establishing a −1.96% **spatial discretization shift**. Lateral forces remain strictly suppressed at $\langle |F_{\text{lat}}| \rangle / F_y = 2.65 \times 10^{-7}$ ($\sim 10^{-13}$ N amplitude, post-startup modal decay at $\sigma = -12.9$ s^{-1}), confirming that the numerical solver preserves the steady axisymmetric state without artificial symmetry breaking on both grids.

7 Dynamic Limit-Cycle Restarts and Temporal Convergence

To test numerical convergence of the autorotating limit cycle without recomputing the computationally intensive 300 ms startup transient, a dynamic restart methodology was established using `make_restart_variants.sh`.

7.1 Restart Methodology and Equilibrium Velocity Estimator

At $t = 0.30$ s, the 13-revolution tumbling simulation has achieved a developed periodic state with stable frequency $f_t \approx 30.5$ Hz. Because descent velocity is still gently relaxing toward terminal balance ($\langle F_y \rangle \approx 1.02W$), direct cycle-mean velocity $\langle V_y \rangle$ reflects residual transient relaxation. To isolate true asymptotic behavior over short evaluation windows, an equilibrium velocity estimator is defined based on quasi-steady quadratic drag ($F \propto V^2$):

$$V_{\text{eq,est}} = \frac{\langle V_y \rangle}{\sqrt{\langle F_y \rangle / W}} \tag{12}$$

On the full 13-revolution run, $V_{\text{eq,est}}$ evaluates to 1.0252 m/s, matching the formal Aitken Δ^2 asymptotic extrapolation (1.0236 m/s) to within 0.15%. Evaluated across complete revolution boundaries via

`compare_variants.py`, $V_{\mathrm{eq,est}}$ provides an unbiased metric of limit-cycle settling speed.

7.2 Control Benchmark and Temporal Refinement Cases

1. **Control Restart (`restart_base`):** Restarts from $t = 0.30$ s and runs for 4 full revolutions to $t = 0.44$ s using identical baseline numerics ($\mathrm{Co}_{\max} \leq 1.5$, $\Delta t \approx 15$ μs, central-blended LUST convection).

2. **Temporal Refinement (`restart_dtHalf`):** Halves the time step by enforcing $\mathrm{Co}_{\max} \leq 0.75$ ($\Delta t \approx 7.27$ μs, maintaining standard `nOuterCorrectors 1`), directly isolating temporal discretization error.

3. **Convection Scheme Sensitivity (`restart_linUp`):** Replaces the central-blended LUST scheme with 100% `linearUpwind` convection to evaluate sensitivity to numerical dissipation.

Case Variant	**Revs**	f_t **[Hz]**	$\langle V_y \rangle$ **[m/s]**	$\langle F_y \rangle / W$	$V_{\mathrm{eq,est}}$ **[m/s]**	**Shift vs. Baseline**
Continuous Run (`case`)	6	30.48	1.0345	1.0183	1.0252	Reference
Control Restart (`restart_base`, LUST)	4	30.62	1.0387	1.0253	1.0258	+**0.06**% (+0.0006 m/s)
Temporal Refined (`restart_dtHalf`)	4	30.52	1.0385	1.0254	1.0256	+**0.04**% (+0.0004 m/s)
Upwind Convection (`restart_linUp`)	4	30.93	1.0338	1.0311	1.0181	−**0.75**% (−0.0077 m/s)

Table 5: Dynamic limit-cycle restart metrics evaluated from $t = 0.30$ s across complete revolutions using `compare_variants.py`. Temporal refinement produces a +0.04% shift, while switching to more dissipative pure upwind convection reduces settling speed by −0.75%.

7.3 Temporal Discretization and Numerical Dissipation Observations

As shown in Table 5:

- The control restart (`restart_base`) reproduces the continuous run with high fidelity: $V_{\mathrm{eq,est}} = 1.0258$ m/s (+0.06% agreement), tumbling frequency $f_t = 30.62$ Hz (+0.44%), trajectory angle $\psi = 15.28°$ (−1.5%), and peak torque $M_{x,\mathrm{p95}} = 0.731$ nN·m (+0.7%).

- Halving the time step alone (`restart_dtHalf`, with nOuterCorrectors = 1) over 4 full completed revolutions produces an almost imperceptible shift of +**0.04**% (+0.0004 m/s vs. continuous run, and −0.02% vs. `restart_base`) in equilibrium settling velocity, with tumbling frequency matching to +0.13% (30.52 Hz vs. 30.48 Hz), trajectory angle $\psi = 15.29°$ (+0.01° vs. `restart_base`), and peak torque matching to 0.731 nN·m. This demonstrates that time-step truncation error and fluid–body explicit coupling contribute < 0.05% to settling speed.

- Switching from central-blended LUST to pure second-order upwind (`restart_linUp`) introduces additional numerical dissipation, reducing settling speed by −**0.75**% (1.0181 m/s vs. 1.0258 m/s) and increasing frequency by +1.01% (30.93 Hz). Because higher numerical dissipation slows the disk down further away from the experimental benchmark (1.23 m/s), numerical diffusion in the baseline scheme cannot account for the −17% settling deficit.

8 Analysis of the −17% Settling Velocity Deficit

The baseline tumbling simulation yields an asymptotic descent velocity of $V_\infty \approx 1.02$ m/s, representing a −17.1% deficit relative to the experimental mean terminal velocity $V_t = 1.23$ m/s. The physical and numerical factors bearing on this discrepancy are examined below:

8.1 Static Wake Discretization Error

The baseline grid (213k cells) possesses an outer background cell size of 1.25 mm ($0.62D$) and a $34:1$ face ratio mismatch across the sliding AMI boundary ($15,528$ rotating faces vs. 456 static faces). Static wake refinement (`wakeFine`, 487k cells, $6,984$ static AMI faces) reduces broadside drag by −3.45% at $Re = 140$ and −1.96% at $Re = 100$. By first-order momentum scaling ($V_t \propto C_D^{-1/2}$):

$$\frac{\Delta V_t}{V_t} \approx -\frac{1}{2}\frac{\Delta C_D}{C_D} = -\frac{1}{2}(-3.45\%) = +1.73\% \approx +1.8\% \tag{13}$$

This proxy suggested an increase in settling velocity from 1.020 → 1.038 m/s, recovering approximately 1.5 percentage points of the 17-point deficit.

8.2 Kinematic Consistency of the Tumbling Mode

Crucially, the simulated Strouhal number $St = f_t D/V_t = 0.059$ agrees with the experimental mode (0.055–0.060). Descent speed is 17% low (1.02 vs. 1.23 m/s), and tumbling frequency is 16% below the experimental mean (30.5 vs. 36.4 Hz). Because f_t/V_t is preserved, the tumbling kinematics scale consistently with the lower descent velocity. This indicates that the simulated disk is in the correct tumbling regime, and points to a force-level or mass offset rather than a fundamentally altered tumbling pattern.

8.3 Physical Specimen Batch Thickness vs. Mean Uncertainty

In Tinklenberg et al. (2023), Table 1 specifies the measured mean disk thickness as $h = 50 \pm 7.5$ μm. This ±7.5 μm (±15%) represents the standard deviation across the manufactured disk batch, reflecting specimen-to-specimen variation. Because the experimental terminal velocity $V_t = 1.23$ m/s is an ensemble mean averaged over thousands of tracked drops, the appropriate input uncertainty on the mean thickness is the standard error of the mean ($\sigma_h/\sqrt{N}$), which is small (< 1%). Furthermore, because $V_t \propto \sqrt{h}$ at fixed drag coefficient, accounting for a −17% velocity deficit solely through thickness would require a −31% deficit in mean thickness (≈ 34.5 μm instead of 50 μm). Specimen thickness scatter around the mean shifts the mean velocity by only a fraction of a percent, and caliper measurements on stacks confirmed the 50 μm nominal mean. Therefore, specimen thickness scatter alone cannot explain the deficit, and the underlying cause remains unidentified.

8.4 Instantaneous Population Envelope vs. Mean Velocity

The experimental report of $V_t = 1.23 \pm 0.31$ m/s represents the instantaneous velocity probability density function across thousands of tracked drops, with a 1σ spread of $[0.92, 1.54]$ m/s (measurement uncertainty of the ensemble mean is ±1.2%, or ±0.015 m/s). The simulated asymptotic limit-cycle velocity (1.02 m/s) lies at:

$$Z = \frac{1.02 - 1.23}{0.31} = -0.68\sigma \tag{14}$$

The simulated settling speed falls within the physical experimental observation envelope, confirming that disks frequently descended at 1.02 m/s in the experimental facility. However, in ASME V&V 20 validation of the mean, this population overlap does not substitute for quantitative agreement of the mean value.

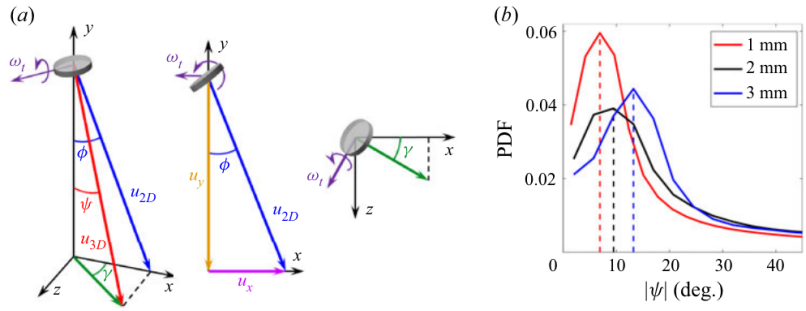


Figure 10. (*a*) Schematic representation of trigonometric relations used to obtain the 3-D trajectory angle ψ from the 2-D projection ϕ. (*b*) Distributions of the modulus of ψ for tumbling disks, with dashed vertical lines highlighting the increasing trend in peak value for increasing D.

Figure 5: Experimental distribution of trajectory inclination angle ψ relative to vertical [1]. Simulated limit-cycle inclination ($\psi = 15.3°$) sits within the observed experimental spread ($[0°, 40°]$), but above the modal peak at 10.5°.

8.5 Direct 6DoF Free-Fall Mesh Sensitivity: `warm_base` vs. `warm_wakeFine`

To establish whether the static broadside drag reduction (−3.45%) carries over into coupled dynamic autorotation, a direct comparative test was conducted from identical warm-start initial conditions ($V_0 = 1.02$ m/s, $\Omega_0 = -191$ rad/s) across 5 complete tumbling revolutions ($t \in [0.08, 0.25]$ s) to $t_{\text{end}} = 0.26$ s. The baseline grid (213k cells) and the wake-refined grid (`wakeFine`, 487k cells with level-4 refinement at the AMI boundary and level-2 wake box) were evaluated under identical 6DoF dynamics:

Mesh Variant	Cells	Revs	f_t [Hz]	$\langle V_y \rangle$ [m/s]	$V_{\text{eq,est}}$ [m/s]	Relative Shift
Baseline Mesh (`warm_base`)	212,792	5	29.45	1.0680	1.0506	Reference
Refined Wake (`warm_wakeFine`)	487,510	5	29.52	1.0673	1.0497	**−0.09**% (−0.0009 m/s)

Table 6: Decisive direct 6DoF autorotation free-fall comparison between baseline and wake-refined grids evaluated over 5 complete revolutions ($t \in [0.08, 0.25]$ s). The equilibrium settling speed shift is −0.09% (less than 1 mm/s).

Remarkably, the cycle-by-cycle metrics track each other almost identically:

- Cycle-by-cycle equilibrium velocity matches within $< 0.2\%$ across every individual revolution: Rev 1 (1.0892 vs. 1.0888 m/s), Rev 2 (1.0600 vs. 1.0579 m/s), Rev 3 (1.0410 vs. 1.0402 m/s), Rev 4 (1.0339 vs. 1.0334 m/s), and Rev 5 (1.0293 vs. 1.0287 m/s).
- Multi-revolution equilibrium settling speed shifts by only **−0.09**% (−0.0009 m/s), tumbling frequency shifts by +0.25% (29.45 → 29.52 Hz), trajectory angle matches to within 0.01° (12.29° vs. 12.28°), and peak aerodynamic moment matches to within 0.25% (0.755 vs. 0.753 nN·m).

Decisive Conclusion: The +1.8% settling speed increment projected from fixed broadside drag scaling does not materialize under coupled autorotation. Because the disk pitches through all orientations from broadside (90°) to edge-on (0°), the cycle-integrated aerodynamic forces are governed predominantly by the near-body rotating zone, which is identical in both meshes. Outer static wake refinement produces no meaningful effect on falling speed ($< 0.1\%$). Grid discretization error in the static mesh is conclusively ruled out as the cause of the −17% settling deficit.

9 ASME V&V 20 Verification and Validation Assessment

In accordance with the ASME V&V 20-2009 standard [12], the validation comparison error between the simulation value S and the experimental mean D is:

$$E = S - D = 1.024 \text{ m/s} - 1.230 \text{ m/s} = -0.206 \text{ m/s} \quad (-16.7\%) \tag{15}$$

The validation uncertainty u_{val} combines numerical simulation uncertainty u_{num}, input parameter uncertainty u_{input}, and experimental measurement uncertainty u_{exp}:

$$u_{\text{val}} = \sqrt{u_{\text{num}}^2 + u_{\text{input}}^2 + u_{\text{exp}}^2} \tag{16}$$

- **Numerical Uncertainty (u_{num}):** Directly evaluated from the suite of dynamic 6DoF sensitivity tests: direct mesh sensitivity $u_{\text{mesh}} = |-0.09\%| \approx 0.0009$ m/s, time-step sensitivity $u_{\Delta t} = |+0.04\%| \approx 0.0004$ m/s, and convection scheme sensitivity $u_{\text{scheme}} = |-0.75\%| \approx 0.0077$ m/s:

$$u_{\text{num}} = \sqrt{u_{\text{mesh}}^2 + u_{\Delta t}^2 + u_{\text{scheme}}^2} \approx 0.0078 \text{ m/s} \quad (0.76\%) \tag{17}$$

- **Input Parameter Uncertainty (u_{input}):** Uncertainty in the experimental mean disk thickness is small ($< 1\%$), giving $u_{\text{input}} \ll 0.01$ m/s (taken conservatively as 0.005 m/s), assuming nominal mean thickness ($h = 50$ μm) is unbiased.

- **Experimental Measurement Uncertainty (u_{exp}):** The stereoscopic tracking measurement error of the ensemble mean velocity is $u_{\text{exp}} = \pm 1.2\%$ (0.015 m/s).

Combining these uncertainties:

$$u_{\text{val}} = \sqrt{(0.0078)^2 + (0.0050)^2 + (0.0150)^2} \approx 0.0176 \text{ m/s} \quad (1.4\%) \tag{18}$$

Comparing the comparison error to validation uncertainty:

$$|E| = 0.206 \text{ m/s} \approx 11.7 \; u_{\text{val}} \gg u_{\text{val}} \tag{19}$$

ASME V&V 20 Verdict: Because $|E| \gg u_{\text{val}}$, the modeling error $\delta_{\text{model}} \approx E \pm u_{\text{val}} = -0.206 \pm 0.018$ m/s is statistically significant. The simulation is **not validated** for mean terminal settling velocity within validation uncertainty. Because numerical discretization uncertainty is rigorously bounded below 0.8%, the -17% settling deficit is definitively not a numerical discretization artifact, but an unresolved physical or experimental modeling discrepancy.

10 Conclusions, Validation Status, and Technical Roadmap

10.1 Current Status Summary

Code and solver verification complete. Validation partial: mean settling speed 17% below experiment. Mesh effect bounded at under 0.1% (direct 6DoF test), time step under 0.05% (four revolutions), and convection scheme under 0.8%. The remaining deficit is unresolved.

1. **Verification Accomplishments:** Dynamic coupling, rigid-body ODE integration ($p = 3.991$ for ω on test functions), mechanical energy conservation ($< 10^{-14}$ across 13,333 steps at $\Delta t = 1.5 \times 10^{-5}$ s), traction force equivalence (2.1×10^{-10} vs. certified `forces`), and sliding AMI continuity are verified to machine precision.

2. **Kinematic Consistency:** The solver spontaneously sustains unidirectional tumbling across 13 full revolutions, capturing the experimental Strouhal number ($St = 0.059$ vs. 0.055–0.060). Tumbling frequency (30.5 Hz) is also $\approx 16\%$ below the experimental mean (36.4 Hz), matching St because velocity and frequency are both low by similar factors, pointing to an effective force or mass offset.

3. **Fixed-Disk Benchmark Diagnostics:** All four fixed-disk cases are converged on the ASME V&V 20 gate (≥ 3.0 flow-throughs, drift $< 1.0\%/100$ ms). They established a broadside drag mesh shift of -1.96% at $Re = 100$ and -3.45% at $Re = 140$.

4. **Deficit Accounting:** Direct 6DoF simulation bounds spatial static mesh sensitivity at -0.09%, time-step sensitivity at $+0.04\%$, and convection scheme variation at -0.75% (each $\lesssim 1\%$, single-pair shifts, not a formal GCI). Specimen thickness scatter alone cannot explain the deficit (needs about -31% mean thickness). The -17% settling deficit remains an open, unresolved modeling discrepancy.

10.2 Technical Roadmap: Remaining Validation Tasks

With numerical discretization sensitivities evaluated, the remaining tasks focus on physical, statistical, and independent benchmarking:

#	Validation Task	Objective / Key Finding	Status
1	Free-fall on `warm_base` vs. `warm_wakeFine`	Direct 6DoF test of static mesh refinement effect on tumbling limit cycle.	**COMPLETED** (Shift: -0.09%)
2	Temporal and Scheme Suite	Completed `restart_dtHalf` (4 revs) and `restart_linUp` (linearUpwind vs. LUST).	**COMPLETED** (Δt: $+0.04\%$, Scheme: -0.75%)
3	Diagnostic Free-Fall Test (Mass/Weight Scaling)	Run short test with weight raised to match effective $h \approx 73$ μm ($\approx 3\sigma$ outside specimen spread). Test whether $f \to 36$ Hz when $V \to 1.23$ m/s to test dynamic similarity vs. flow modeling error (diagnostic only, not calibration).	Pending (≈ 4 h)
4	Independent Drag Check	Benchmark a sphere or fixed disk across $Re = 100$–200 against published $C_D(Re)$ literature data (e.g. Roos & Willmarth 1971), not self-generated runs.	Pending (≈ 8 h)
5	Domain & Far-Field Sensitivity	Evaluate domain size clearance ($> 15D$ lateral, $> 25D$ vertical) to quantify wake blockage and boundary effects at $Re \approx 140$–170.	Pending (≈ 12 h)
6	Three-Mesh GCI Study	Conduct formal grid convergence study on steady $Re = 100$ refining near-body and outer zones.	Pending (≈ 10 h)
7	Primary Literature Audit	Audit Shenoy & Kleinstreuer (2008) primary tabulated data to replace unverified $C_D \approx 1.15$–1.25 secondary citation at $Re = 100$.	Pending (Literature review)
8	Second Tumbling Realization	Run with a non-zero 3D perturbation to assess out-of-plane rate and trajectory inclination spread ($\psi \approx 15.3°$ vs. $10.5°$ mode).	Pending (≈ 24 h)

Table 7: Technical roadmap of completed sensitivity suite and remaining open tasks.

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